BrightLight Tutorials
Data Analytics
Exercise 1 : SQL Fundamentals
Database : employees_db

1 SELECT Statement

Write a SQL query to retrieve all columns from the employees table

```
Select *
From employees;
```

2 SELECT DISTINCT Statement

Write a SQL query to find all the unique departments in the employees table.

```
Select first_name,
       Last_name,
Distinct Department
From employees;
```

Department
IT
HR
FINANCE
MARKETING

3 ORDER BY Statement

Write a SQL query to retrieve all employees first and last names, ordered by salary in descending order

```
Select First_name,
       Last_name,
       Salary
From employees
Order By Salary Desc;
```

3

First_name	Last_name	Salary
Emily	Davis	62000
Mike	Johnson	60000
Robert	Wilson	59000
John	Doe	55000
Daniel	Clark	53000
Sarah	Brown	53000
David	White	52000
Jessica	Moore	51000
Laura	Hall	50000
Jane	Smith	48000

4 LIMIT Statement

Write a SQL query to retrieve the top 5 highest-paid employees.

```
Select  First_name,
        Last_name,
        Salary
FROM employees
ORDER BY Salary Desc,
Limit 5;
```

First_name	Last_name	Salary
Emily	Davis	62000
Mike	Johnson	60000
Robert	Wilson	59000
John	Doe	55000
Daniel	Clark	53000

5 Where Statement
Write a SQL query to find employees who work in the IT department

```
Select  First_name,
        Last_name,
        Department
FROM    employees
Where   department = 'IT';
```

First Last_name	Last name	Department
John	Doe	IT
Sarah	Brown	IT
Emily	Davis	IT
Laura	Hall	IT

6 AND Statement
Write a SQL query to find employees who work in the Finance department AND have a salary greater than 58 000

```
Select  First_name,
        Last_name,
        Department,
        Salary
From    employees
Where   department = 'Finance' AND 'Salary' > 58 000;
```

First_name	Last name	Department	Salary
Mike	Johnson	Finance	60 000
Robert	Wilson	Finance	59 000

7 OR Statement

Write a SQL to find employees who work in the HR department OR the Marketing department.

```
Select First_name,
       Last_name,
       Department
From employees
Where department = 'Marketing' OR 'HR';
```

First_name	Last_name	Department
Jane	Smith	HR
David	White	Marketing
Jessica	Moore	HR
Daniel	clark	Marketing

8 NOT Statement

Write a SQL query of employees who do not work in the IT department.

```
Select First_name,
       Last_name,
       department
From employees
Where NOT department = 'IT';
```

First_name	Last_name	Department
Jane	Smith	HR
Mike	Johnson	Finance
David	White	Marketing
Robert	Wilson	Finance
Jessica	Moore	HR
Daniel	clark	Marketing

9 In Statement
Write a SQL query to find employees who are in the HR, IT, or Finance department

```
Select First_name,
       Last_name,
       Department
From employees
Where department IN 'HR', 'IT', 'Finance';
```

First_Name	Last_name	Department
John	Doe	IT
Jane	Smith	HR
Mike	Johnson	Finance
Sarah	Brown	IT
Emily	Davis	IT
Robert	Wilson	Finance
Jessica	Moore	HR
Laura	Hall	IT

10 Combining Conditions
Write a SQL query to find the employees who are in the IT department, have a salary greater than 50 000 and are located in New York

```
Select First_name,
       Last_name,
       department,
       Salary, City
From employees
Where department = 'IT' AND Salary > 50 000
      AND City = 'New York';
```

Salary	First name	Last name	Department	City
55000	John	Doe	IT	New York
53000	Sarah	Brown	IT	New York

11 Combining WHERE, AND and ORDER BY
Write a SQL query to retrieve the first and last names of employees who work in the Finance or Marketing department, earn more than 52 000 and order the results by Salary in descending order

```
Select First_name,
       Last_name,
       Department,
       Salary
from employees
Where Department = 'Finance' OR Department = 'Marketing'
AND Salary > 52 000,
GROUP BY Salary Desc;
```

First_name	Last_name	Department	Salary
Mabel	Johnson	Finance	60 000
Robert	Wilson	Finance	59000
Daniel	Clark	Marketing	53 000

12 Combining SELECT DISTINCT, WHERE and IN
Write a SQL query to find all the unique cities where employees work, excluding those in the IT and HR departments
```
Select Distinct City
from employees
Where department NOT IN 'IT', 'HR';
```

City
Los Angeles
San Francisco
Houston
Chicago

13 Combining WHERE, NOT, AND and ORDER BY
Write a SQL query to retrieve employees who are NOT in the Finance department, have a salary greater than 50 000, and order the results by hire date in ascending order.

```
Select First_name,
       Last_name,
       Salary,
       department
From employees
Where department NOT 'FINANCE'
AND Salary > 50 000
ORDER BY hire_date ASC;
```

First_name	Last_name	Department	Salary
John	Doe	IT	55 000
~~Jane~~	~~Smith~~	~~HR~~	~~48 000~~
Sarah	Brown	IT	53 000
David	White	Marketing	52 000
Emily	Davis	IT	62 000
Jessica	Moore	HR	51 000
Daniel	Clark	Marketing	53 000
~~Laura~~	~~Hall~~	~~IT~~	

14 Combining WHERE, OR, IN, and LIMIT

Write a SQL query to find the first 3 employees who work in either Chicago or Los Angeles and belong to the IT or Marketing department.

```
Select First_name,
       Last_name,
       Department,
       City
From employees
Where City IN ('Chicago' OR 'Los Angeles')
AND department = 'IT' OR department = 'Marketing'
Limit 3;
```

First name	Last name	Department	City
Emily	Davis	IT	Chicago
Daniel	Clark	Marketing	Chicago

Exercise 2

SQL Exercise 'Aggregate functions and Grouping

1 Count () function

Write a SQL query to find the total number of employees in the company.

```
Select COUNT (*) As total_employees
From employees;
```

Total_employees
10